

$$n \mid \phi(a^n - 1)$$

$$\text{Ord}_{a^n-1}(a) = n$$

$$a^n \equiv 1 \pmod{a^n-1}$$

$$a^k$$

$$5^{2^{l-3}} = (2^2 + 1)^{2^{l-3}}$$

$$= \sum_{k=0}^{2^{l-3}} 2^{2k} \binom{2^{l-3}}{k}$$

if

$$(a_1, \dots, a_k) = 1$$

$$5^{2^{l-3}} \equiv 2^{l-1} + 1 \pmod{2^l}$$

$$\phi(2^l) = 2^l - 2^{l-1} = 2^{l-1}$$

$$5^{2^l} \equiv 1$$

$$(5^{2^{l-1}} - 1) (5^{2^{l-1}} + 1) \equiv 0 \pmod{2^l}$$

$$(5^{2^{l-3}} - 1) (5^{2^{l-3}} + 1) (5^{2^{l-2}} + 1)$$

$$(5^{2^{l-1}} + 1) = Q$$

$$5^{2^k} + 1 \equiv 1^{2^k} + 1$$

$$\equiv 2 \pmod{4}$$

$$2^l \mid Q$$

$$\text{also } 4 \nmid 5^{2^k} + 1$$

$$\text{ii } 2 \mid 5^{2^k} + 1$$

$$\Rightarrow 2^{l-3} \mid 5^{2^{l-3}} - 1$$

$$5^{2^{l-3}} - 1 = 2^{l-3} \cdot d$$

$$5^{2^{l-3}} = 2^{l-3} \cdot d + 1$$

$$2^{l-3} \cdot d + 1 \stackrel{?}{\equiv} 2^{l-3} + 1 \pmod{2^l}$$

$$2^{l-3} \cdot d \stackrel{?}{\equiv} 2^{l-3} \pmod{2^l}$$

$$d \stackrel{?}{\equiv} 1 \pmod{8}$$

$$d \cdot 2^{l-3}$$